

Rutuja Patel

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EDUCATION

W. P. Carey School of Business at Arizona State University

Master of Science in Business Analytics.

Tempe, AZ

Dec 2025

Relevant Coursework: AI and Data Analytics, Descriptive and Predictive Analytics, Analytics of Unstructured Data, Machine Learning in Business, Advanced Marketing Analytics, Enterprise Data Analytics, Operations Planning/Execution, Operations & Supply Chain Management, Quantitative Risk Management, Strategic Procurement, Logistics in the Supply Chain, Business Process Analytics

Gujarat Technological University

Bachelor of Engineering in Electronics and Communication.

Ahmedabad, GJ

July 2020

EXPERIENCE

Applied Analyst (Capstone + Internship)

DST Investment Advisors

Aug 2025 – Present

Scottsdale, AZ

- Build data mining and web-scraping pipelines (Playwright, Python) to extract and structure attorney lead data from multiple government sources into clean, analysis-ready data products, applying data governance, quality-control, and standardization best practices.
- Automate the end-to-end data-oriented programming workflow, significantly reducing manual research time and ensuring consistent, on-time data delivery for stakeholder reporting.

ERP Data Assistant

Hexaplast Industries (Electronics Manufacturing)

Jan 2024 – May 2024

Ahmedabad, GJ

- Queried and managed structured data using SQL, alongside Python and Excel, in an ETL workflow to clean, transform, and consolidate multi-source data (office documents, sales contracts, real-time order data) supporting ERP implementation and cross-departmental reporting.

Business Operations Trainee

Hexaplast Industries (Electronics Manufacturing)

May 2023 – Dec 2023

Ahmedabad, GJ

- Completed cross-departmental training across sales, inventory, machine building, and electronics to build end-to-end operational and manufacturing knowledge ahead of specializing in data analytics.

PROJECTS

Hospital Readmission of Diabetic Patients

ASU Data Science Lab

Aug 2025 – Dec 2025

- Applied machine learning and data modeling techniques to a 101,766-record diabetic patient dataset from 130 U.S. hospitals to predict 30-day readmission risk, building and evaluating Logistic Regression, Random Forest, and XGBoost models (0.672 AUC-ROC).
- Validated model robustness via 5-fold stratified cross-validation and supported responsible data governance by auditing 50 data fields and excluding race/gender from model inputs to mitigate bias.
- Presented methodology and findings to a cross-disciplinary audience of data scientists, healthcare professionals, students, and librarians; published full technical documentation on a public portfolio site for transparency and reproducibility.

Diabetes Readmission Risk Assistant (RAG Application)

Jan 2026 – Feb 2026

- Extended the readmission model into a proof-of-concept healthcare decision-support application, designing a retrieval-augmented generation pipeline over 46 clinical guideline documents (ADA, WHO, CDC, Mayo Clinic) chunked into 2,456 embeddings to drive evidence-based decision-making.
- Integrated a live XGBoost risk score with an LLM (Llama-3.1-8B) via Streamlit to deliver personalized, citation-backed care recommendations.

Inventory Optimization Using Computer Vision

Mar 2025 – May 2025

- Partnered with ASU's Pitchforks dining hall to automate a 6-8 hour manual inventory count, fine-tuning a YOLOv8 object-detection model via transfer learning on a self-collected 428-image dataset (96.8% precision, 0.973 mAP@0.5).
- Diagnosed a multi-object detection failure mode and resolved it using synthetic data augmentation, limiting the mAP@0.5 drop to just 0.011.

Fast Fashion vs. Slow Fashion (Yelp Review Analysis)

Jan 2025 – Mar 2025

- Compared customer sentiment across retailers by cleaning and joining 5 raw Yelp datasets (~10M records) into a 65,729-review, 3,100-business dataset spanning 11 U.S. metros, staging and processing the data on AWS (S3) to manage its scale.
- Ran dual NLP sentiment pipelines (VADER + DistilBERT) and applied TF-IDF/LDA topic modeling across 6 complaint categories, building an analytical framework with key metrics to inform business strategy recommendations.

Reviewer Social Network Analysis

Feb 2026 – Apr 2026

- Investigated whether Yelp's "Elite" reviewer status reflects genuine social influence by constructing a social graph of 46,236 reviewers and 135,151 friendships, applying graph-based data modeling techniques.
- Found that Veteran Elite reviewers were ~44x more socially connected than single-review users, using machine learning-based k-means geospatial clustering across 11 U.S. metros.

SKILLS

Programming & Data-Oriented Languages: Python, SQL, HTML, CSS, JavaScript

Cloud & Data Engineering: AWS (S3), ETL/ELT Pipelines, Data Pipelines, Data Modeling, Data Governance & Quality Standardization

Analytics & Visualization: Excel, Power BI, Tableau, Performance Dashboards

Machine Learning & Statistics: Statistical Analysis, scikit-learn, XGBoost, Random Forest, Logistic Regression

AI & Generative AI: Retrieval-Augmented Generation (RAG), LLMs (Llama-3.1-8B), Sentence-Transformers, ChromaDB, Streamlit

Deep Learning & Computer Vision: YOLOv8, OpenCV, FastSAM, DistilBERT, Transfer Learning

NLP, Data Mining & Automation: Sentiment Analysis (VADER), Topic Modeling (TF-IDF/LDA), NetworkX (Graph Analysis), Geospatial Clustering, Playwright (Web Automation)